Exercise 14.3 SOLUTION

Q1: Compute the cross product $\underline{a}\times\underline{b}$ and $\underline{b}\times\underline{a}$. Check your answer by showing that each $\underline{a}$ and $\underline{b}$ are perpendicular to $\underline{a}\times\underline{b}$ and $\underline{b}\times\underline{a}$.

i. $\underline{a}=2\underline{i}+\underline{j}-\underline{k}$ and $\underline{b}=\underline{i}-\underline{j}+\underline{k}$

Sol: $\underline{a}=2\underline{i}+\underline{j}-\underline{k}$ and $\underline{b}=\underline{i}-\underline{j}+\underline{k}$

$$\underline{a}\times\underline{b}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\2&1&-1\\1&-1&1\end{vmatrix}=\underline{i}(1-1)-\underline{j}(2+1)+\underline{k}(-2-1)=0\underline{i}-3\underline{j}-3\underline{k}$$

$$\underline{b}\times\underline{a}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\1&-1&1\\2&1&-1\end{vmatrix}=\underline{i}(1-1)-\underline{j}(-1-2)+\underline{k}(1+2)=0\underline{i}+3\underline{j}+3\underline{k}$$

نوٹ:
اگر $\underline{u}$ اور $\underline{v}$ دو ویکٹر کا dot product صفر ہو گا تو دونوں ویکٹر Perpendicular ہوں گے اور اگر cross product صفر ہو گا تو parallel ہوں گے

Now $\underline{a}.(\underline{a}\times\underline{b})=(2\underline{i}+\underline{j}-\underline{k})\cdot(-3\underline{j}-3\underline{k})=(2)(0)+(1)(-3)+(-1)(-3)=-3+3=0$

So, $\underline{a}$ and $\underline{a}\times\underline{b}$ are perpendicular.

$\underline{a}.(\underline{b}\times\underline{a})=(2\underline{i}+\underline{j}-\underline{k})\cdot(3\underline{j}+3\underline{k})=(2)(0)+(1)(3)+(-1)(3)=0+3-3=0$

So, $\underline{a}$ and $\underline{b}\times\underline{a}$ are perpendicular.

$\underline{b}.(\underline{a}\times\underline{b})=(\underline{i}-\underline{j}+\underline{k})\cdot(-3\underline{j}-3\underline{k})=(1)(0)+(-1)(-3)+(1)(-3)=3-3=0$

So, $\underline{b}$ and $\underline{a}\times\underline{b}$ are perpendicular.

$\underline{b}.(\underline{b}\times\underline{a})=(\underline{i}-\underline{j}+\underline{k})\cdot(3\underline{j}+3\underline{k})=(1)(0)+(-1)(3)+(1)(3)=3-3=0$

So, $\underline{b}$ and $\underline{b}\times\underline{a}$ are perpendicular.

ii. $\underline{a}=\underline{i}+3\underline{j}+2\underline{k}$ and $\underline{b}=2\underline{i}-\underline{j}+\underline{k}$

Sol: $\underline{a}=\underline{i}+3\underline{j}+2\underline{k}$ and $\underline{b}=2\underline{i}-\underline{j}+\underline{k}$

$$\underline{a}\times\underline{b}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\1&3&2\\2&-1&1\end{vmatrix}=\underline{i}(3+2)-\underline{j}(1-4)+\underline{k}(-1-6)=5\underline{i}+3\underline{j}-7\underline{k}$$

$$\underline{b}\times\underline{a}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\2&-1&1\\1&3&2\end{vmatrix}=\underline{i}(-2-3)-\underline{j}(4-1)+\underline{k}(6+1)=-5\underline{i}-3\underline{j}+7\underline{k}$$

Now $\underline{a}.(\underline{a}\times\underline{b})=(\underline{i}+3\underline{j}+2\underline{k})\cdot(5\underline{i}+3\underline{j}-7\underline{k})=(1)(5)+(3)(3)+(2)(-7)=5+9-14=0$

So, $\underline{a}$ and $\underline{a}\times\underline{b}$ are perpendicular.

$\underline{a}.(\underline{b}\times\underline{a})=(\underline{i}+3\underline{j}+2\underline{k})\cdot(-5\underline{i}-3\underline{j}+7\underline{k})=(1)(-5)+(3)(-3)+(2)(7)=-5-9+14=0$

So, $\underline{a}$ and $\underline{b}\times\underline{a}$ are perpendicular.

$\underline{b}.(\underline{a}\times\underline{b})=(2\underline{i}-\underline{j}+\underline{k})\cdot(5\underline{i}+3\underline{j}-7\underline{k})=(2)(5)+(-1)(3)+(2)(-7)=10-3-7=0$

So, $\underline{b}$ and $\underline{a}\times\underline{b}$ are perpendicular.

$\underline{b}.(\underline{b}\times\underline{a})=(2\underline{i}-\underline{j}+\underline{k})\cdot(-5\underline{i}-3\underline{j}+7\underline{k})=(2)(-5)+(-1)(-3)+(1)(7)=-10+3+7=0$

So, $\underline{b}$ and $\underline{b}\times\underline{a}$ are perpendicular.

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iii. $\underline{a} = 2\underline{i} - 2\underline{j} + \underline{k}$ and $\underline{b} = -\underline{i} + \underline{j} + 3\underline{k}$

Sol: $\underline{a} = 2\underline{i} - 2\underline{j} + \underline{k}$ and $\underline{b} = -\underline{i} + \underline{j} + 3\underline{k}$

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -2 & 1 \\ -1 & 1 & 3 \end{vmatrix} = \underline{i}(-6-1) - \underline{j}(6+1) + \underline{k}(2-2) = -7\underline{i} - 7\underline{j} + 0\underline{k}$$

$$\underline{b} \times \underline{a} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{vmatrix} = \underline{i}(1+6) - \underline{j}(-1-6) + \underline{k}(2-2) = 7\underline{i} + 7\underline{j} + 0\underline{k}$$

Now $\underline{a}.(\underline{a} \times \underline{b}) = (2\underline{i} - 2\underline{j} + \underline{k}) \cdot (-7\underline{i} - 7\underline{j} + 0\underline{k}) = (2)(-7) + (-2)(-7) + (1)(0) = -14 + 14 + 0 = 0$

So, $\underline{a}$ and $\underline{a} \times \underline{b}$ are perpendicular.

$\underline{b}.(\underline{a} \times \underline{b}) = (-1\underline{i} + 1\underline{j} + 3\underline{k}) \cdot (-7\underline{i} - 7\underline{j} + 0\underline{k}) = (-1)(-7) + (1)(-7) + (3)(0) = 7 - 7 + 0 = 0$

So, $\underline{b}$ and $\underline{a} \times \underline{b}$ are perpendicular.

$\underline{a}.(\underline{b} \times \underline{a}) = (2\underline{i} - 2\underline{j} + \underline{k}) \cdot (7\underline{i} + 7\underline{j} + 0\underline{k}) = (2)(7) + (-2)(7) + (1)(0) = 14 - 14 + 0 = 0$

So, $\underline{a}$ and $\underline{b} \times \underline{a}$ are perpendicular.

$\underline{b}.(\underline{b} \times \underline{a}) = (-1\underline{i} + 1\underline{j} + 3\underline{k}) \cdot (7\underline{i} + 7\underline{j} + 0\underline{k}) = (-1)(7) + (1)(7) + (3)(0) = 7 + 7 + 0 = 0$

So, $\underline{b}$ and $\underline{b} \times \underline{a}$ are perpendicular.

iv. $\underline{a} = -4\underline{i} + \underline{j} - 2\underline{k}$ and $\underline{b} = 2\underline{i} + \underline{j} + \underline{k}$

Sol: $\underline{a} = -4\underline{i} + \underline{j} - 2\underline{k}$ and $\underline{b} = 2\underline{i} + \underline{j} + \underline{k}$

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -4 & 1 & -2 \\ 2 & 1 & 1 \end{vmatrix} = \underline{i}(1+2) - \underline{j}(-4+4) + \underline{k}(-4-2) = 3\underline{i} - 0\underline{j} - 6\underline{k}$$

$$\underline{b} \times \underline{a} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & 1 \\ -4 & 1 & -2 \end{vmatrix} = \underline{i}(-2-1) - \underline{j}(-4+4) + \underline{k}(2+4) = -3\underline{i} - 0\underline{j} + 6\underline{k}$$

Now $\underline{a}.(\underline{a} \times \underline{b}) = (-4\underline{i} + \underline{j} - 2\underline{k}) \cdot (3\underline{i} - 0\underline{j} - 6\underline{k}) = (-4)(3) + (1)(-0) + (-2)(-6) = -12 - 0 + 12 = 0$

So, $\underline{a}$ and $\underline{a} \times \underline{b}$ are perpendicular.

$\underline{b}.(\underline{a} \times \underline{b}) = (2\underline{i} + \underline{j} + \underline{k}) \cdot (3\underline{i} - 0\underline{j} - 6\underline{k}) = (2)(3) + (1)(-0) + (1)(-6) = 6 - 0 - 6 = 0$

So, $\underline{b}$ and $\underline{a} \times \underline{b}$ are perpendicular.

$\underline{a}.(\underline{b} \times \underline{a}) = (2\underline{i} + \underline{j} + \underline{k}) \cdot (-3\underline{i} - 0\underline{j} + 6\underline{k}) = (2)(-3) + (1)(-0) + (1)(6) = -6 - 0 + 6 = 0$

So, $\underline{a}$ and $\underline{b} \times \underline{a}$ are perpendicular.

$\underline{b}.(\underline{b} \times \underline{a}) = (2\underline{i} + \underline{j} + \underline{k}) \cdot (-3\underline{i} - 0\underline{j} + 6\underline{k}) = (2)(-3) + (1)(-0) + (1)(6) = -6 - 0 + 6 = 0$

So,

$\underline{b}$ and $\underline{b} \times \underline{a}$ are perpendicular.

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Q2. **Find a unit vector perpendicular to the plane containing $\underline{a}$ and $\underline{b}$. Also find sine of the angle between them:**

i. $\underline{a}=\underline{i}+6\underline{j}-3\underline{k}, \underline{b}=2\underline{i}+\underline{j}+3\underline{k}$

Sol: $\underline{a}=\underline{i}+6\underline{j}-3\underline{k}, \underline{b}=2\underline{i}+\underline{j}+3\underline{k}$

Given in component form:

$$\underline{a}\times\underline{b}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 1 & 6 & -3\\ 2 & 1 & 3\end{vmatrix}=\underline{i}(18+3)-\underline{j}(3+6)+\underline{k}(1-12)=21\underline{i}-9\underline{j}-11\underline{k}$$

$$|\underline{a}\times\underline{b}|=\sqrt{(21)^2+(-9)^2+(-11)^2}=\sqrt{442+81+121}=\sqrt{643}$$

Unit vector perpendicular to the plane:

$$\hat{n}=\frac{\underline{a}\times\underline{b}}{|\underline{a}\times\underline{b}|}=\frac{1}{\sqrt{643}}(21\underline{i}-9\underline{j}-11\underline{k})$$

Compute $|\underline{a}|$ and $|\underline{b}|$:

$$|\underline{a}|=\sqrt{(1)^2+(6)^2+(-3)^2}=\sqrt{1+36+9}=\sqrt{46}$$

$$|\underline{b}|=\sqrt{(2)^2+(1)^2+(3)^2}=\sqrt{4+1+9}=\sqrt{14}$$

Sine of the angle between $\underline{a}$ and $\underline{b}$:

$$\sin\theta=\frac{|\underline{a}\times\underline{b}|}{|\underline{a}||\underline{b}|}=\frac{\sqrt{643}}{\sqrt{46}\sqrt{14}}=\frac{\sqrt{643}}{\sqrt{644}}$$

ii. $\underline{a}=-\underline{i}-\underline{j}-\underline{k}, \underline{b}=2\underline{i}-3\underline{j}+4\underline{k}$

Sol: $\underline{a}=-\underline{i}-\underline{j}-\underline{k}, \underline{b}=2\underline{i}-3\underline{j}+4\underline{k}$

$$\underline{a}\times\underline{b}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ -1 & -1 & -1\\ 2 & -3 & 4\end{vmatrix}=\underline{i}(-4-3)-\underline{j}(-4+2)+\underline{k}(3+2)=-7\underline{i}+2\underline{j}+5\underline{k}$$

$$|\underline{a}\times\underline{b}|=\sqrt{(-7)^2+(2)^2+(5)^2}$$

$$=\sqrt{49+4+25}=\sqrt{78}$$

Unit vector perpendicular to the plane:

$$\hat{n}=\frac{\underline{a}\times\underline{b}}{|\underline{a}\times\underline{b}|}=\frac{1}{\sqrt{78}}(-7\underline{i}+2\underline{j}+5\underline{k})$$

Compute $|\underline{a}|$ and $|\underline{b}|$:

$$|\underline{a}|=\sqrt{(-1)^2+(-1)^2+(-1)^2}=\sqrt{1+1+1}=\sqrt{3}$$

$$|\underline{b}|=\sqrt{(2)^2+(-3)^2+(4)^2}=\sqrt{4+9+16}=\sqrt{29}$$

Sine of the angle between $\underline{a}$ and $\underline{b}$:

$$\sin\theta=\frac{|\underline{a}\times\underline{b}|}{|\underline{a}||\underline{b}|}=\frac{\sqrt{78}}{\sqrt{3}\sqrt{29}}=\frac{\sqrt{78}}{\sqrt{3\times29}}=\frac{\sqrt{78}}{\sqrt{87}}$$

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iii. $\underline{a}=\underline{i}+\underline{j}+\underline{k},\underline{b}=\underline{i}-\underline{j}-\underline{k}$

Sol: $\underline{a}=\underline{i}+\underline{j}+\underline{k},\underline{b}=\underline{i}-\underline{j}-\underline{k}$

$$\underline{a}\times\underline{b}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 1 & 1 & 1\\ 1 & -1 & -1\end{vmatrix}$$

$$=\underline{i}(-1+1)-\underline{j}(-1-1)+\underline{k}(-1-1)=0\underline{i}+2\underline{j}-2\underline{k}$$

$$|\underline{a}\times\underline{b}|=\sqrt{(0)^2+(2)^2+(-2)^2}=\sqrt{0+4+4}=\sqrt{8}=2\sqrt{2}$$

Unit vector perpendicular to the plane:

$$\hat{n}=\frac{\underline{a}\times\underline{b}}{|\underline{a}\times\underline{b}|}=\frac{1}{2\sqrt{2}}\left(0\underline{i}+2\underline{j}-2\underline{k}\right)$$

$$\hat{n}=\frac{1}{\sqrt{2}}\left(0\underline{i}+\underline{j}-\underline{k}\right)$$

Compute $|\underline{a}|$ and $|\underline{b}|$:

$$|\underline{u}|=\sqrt{(-3)^2+(2)^2+(5)^2}$$

$$=\sqrt{9+4+25}=\sqrt{38}$$

$$|\underline{b}|=\sqrt{(1)^2+(-1)^2+(-1)^2}=\sqrt{1+1+1}=\sqrt{3}$$

Sine of the angle between $\underline{a}$ and $\underline{b}$:

$$\sin\theta=\frac{|\underline{a}\times\underline{b}|}{|\underline{a}||\underline{b}|}=\frac{2\sqrt{2}}{\sqrt{3}\sqrt{3}}=\frac{2\sqrt{2}}{3}$$

iv. $\underline{a}=5\underline{i}+\underline{j}-3\underline{k},\underline{b}=-2\underline{i}+4\underline{j}+\underline{k}$

Sol: $\underline{a}=5\underline{i}+\underline{j}-3\underline{k},\underline{b}=-2\underline{i}+4\underline{j}+\underline{k}$

$$\underline{a}\times\underline{b}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 5 & 1 & -3\\ -2 & 4 & 1\end{vmatrix}=\underline{i}(1+12)-\underline{j}(5-6)+\underline{k}(20+2)=13\underline{i}+\underline{j}+22\underline{k}$$

$$|\underline{a}\times\underline{b}|=\sqrt{(13)^2+(1)^2+(22)^2}=\sqrt{169+1+484}=\sqrt{654}$$

Unit vector perpendicular to the plane:

$$\hat{n}=\frac{\underline{a}\times\underline{b}}{|\underline{a}\times\underline{b}|}=\frac{1}{\sqrt{654}}\left(13\underline{i}+\underline{j}+22\underline{k}\right)$$

Compute $|\underline{a}|$ and $|\underline{b}|$:

$$|\underline{a}|=\sqrt{(5)^2+(1)^2+(-3)^2}=\sqrt{25+1+9}=\sqrt{35}$$

$$|\underline{b}|=\sqrt{(-2)^2+(4)^2+(1)^2}=\sqrt{4+16+1}=\sqrt{21}$$

Sine of the angle between $\underline{a}$ and $\underline{b}$:

$$\sin\theta=\frac{|\underline{a}\times\underline{b}|}{|\underline{a}||\underline{b}|}=\frac{\sqrt{654}}{\sqrt{35}\sqrt{21}}=\frac{\sqrt{654}}{\sqrt{35\times21}}=\frac{\sqrt{654}}{\sqrt{735}}$$

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Q3. Find the area of the triangle, formed by the points P, Q and R.

i. $P(2,3,5); Q(1,2,0); R(4,1,2)$

Sol: First, find the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (1-2)\underline{i} + (2-3)\underline{j} + (0-5)\underline{k} = -\underline{i} - \underline{j} - 5\underline{k}$$

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP} = (4-2)\underline{i} + (1-3)\underline{j} + (2-5)\underline{k} = 2\underline{i} - 2\underline{j} - 3\underline{k}$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & -1 & -5 \\ 2 & -2 & -3 \end{vmatrix} = \underline{i}(3-10) - \underline{j}(3+10) + \underline{k}(2+2) = -7\underline{i} - 13\underline{j} + 4\underline{k}$$

$$\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \sqrt{(-7)^2 + (-13)^2 + (4)^2} = \sqrt{49+169+16} = \sqrt{234} = \sqrt{9 \times 26} = 3\sqrt{26}$$

Finally, calculate the area of the triangle:

Area $\frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{3\sqrt{26}}{2}$ square units.

ii. $P(0,0,1); Q(2,-1,2); R(-1,3,2)$

Sol: First, find the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (2-0)\underline{i} + (-1-0)\underline{j} + (2-1)\underline{k} = 2\underline{i} - \underline{j} + \underline{k}$$

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP} = (-1-0)\underline{i} + (3-0)\underline{j} + (2-1)\underline{k} = -\underline{i} + 3\underline{j} + \underline{k}$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -1 & 1 \\ -1 & 3 & 1 \end{vmatrix} = \underline{i}(-1-3) - \underline{j}(2+1) + \underline{k}(6-1) = -4\underline{i} - 3\underline{j} + 5\underline{k}$$

$$\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \sqrt{(-4)^2 + (-3)^2 + (5)^2} = \sqrt{16+9+25} = \sqrt{50}$$

Area of triangle $= \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{1}{2}\sqrt{50} = \frac{1}{2}\sqrt{25 \times 2} = \frac{1}{2} \times 5\sqrt{2} = \frac{5\sqrt{2}}{2}$ square units.

Q4. Find the area of a parallelogram, whose vertices are:

i. $A(1,1,1); B(4,2,3); C(5,6,7); D(2,5,5)$

Sol: First, calculate the vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (4-1)\underline{i} + (2-1)\underline{j} + (3-1)\underline{k} = 3\underline{i} + \underline{j} + 2\underline{k}$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = (2-1)\underline{i} + (5-1)\underline{j} + (5-1)\underline{k} = \underline{i} + 4\underline{j} + 4\underline{k}$$

$$\overrightarrow{AB} \times \overrightarrow{AD} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & 1 & 2 \\ 1 & 4 & 4 \end{vmatrix} = \underline{i}(4-8) - \underline{j}(12-2) + \underline{k}(12-1) = -4\underline{i} - 10\underline{j} + 11\underline{k}$$

$$\left|\overrightarrow{AB} \times \overrightarrow{AD}\right| = \sqrt{(-4)^2 + (-10)^2 + (11)^2} = \sqrt{16+100+121} = \sqrt{237}$$

The area of the parallelogram is $\sqrt{237}$ square units.

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ii. $A(4,5,6); B(1,3,2); C(-2,0,1); D(1,2,5)$

Sol: First, calculate the vectors AB and AD:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (1-4)\underline{i} + (3-5)\underline{j} + (2-6)\underline{k} = -3\underline{i} - 2\underline{j} - 4\underline{k}$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = (1-4)\underline{i} + (2-5)\underline{j} + (5-6)\underline{k} = -3\underline{i} - 3\underline{j} - \underline{k}$$

$$\overrightarrow{AB} \times \overrightarrow{AD} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -3 & -2 & -4 \\ -3 & -3 & -1 \end{vmatrix} = \underline{i}(2-12) - \underline{j}(3-12) + \underline{k}(9-6) = -10\underline{i} - 9\underline{j} + 3\underline{k}$$

$$\left|\overrightarrow{AB} \times \overrightarrow{AD}\right| = \sqrt{(-10)^2 + (9)^2 + (3)^2} = \sqrt{100+81+9} = \sqrt{190}$$

The area of the parallelogram for is $\sqrt{190}$ square units.

Q5. If the cross product of the vectors $\underline{u} = 7\underline{i} - 4\underline{j} + 5\underline{k}$ and $\underline{v} = a\underline{i} - b\underline{j} + 3\underline{k}$ is zero, then find values of a and b

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Sol: $\underline{u} = 7\underline{i} - 4\underline{j} + 5\underline{k}, \underline{v} = a\underline{i} - b\underline{j} + 3\underline{k}$

$$\underline{u} \times \underline{v} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 7 & -4 & 5 \\ a & -b & 3 \end{vmatrix} = \underline{i}(-12+5b) - \underline{j}(21-5a) + \underline{k}(-7b+4a)$$

Given that $\underline{u} \times \underline{v} = 0$

$$(-12+5b)\underline{i} - (21-5a)\underline{j} + (-7b+4a)\underline{k} = 0 \approx 0\underline{i} + 0\underline{j} + 0\underline{k}$$

$-12+5b=0$ and $21-5a=0$

$5b=12$ and $5a=21$

$b=\frac{12}{5}$ and $a=\frac{21}{5}$

Q6. Which vectors, if any, are perpendicular or parallel?

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i. $\underline{u} = 5\underline{i} - \underline{j} + \underline{k}; \underline{v} = \underline{j} - 5\underline{k}; \underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k}$

Sol: $\underline{u} = 5\underline{i} - \underline{j} + \underline{k}; \underline{v} = \underline{j} - 5\underline{k}; \underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k}$

$$\underline{u} \cdot \underline{v} = (5\underline{i} - \underline{j} + \underline{k})(\underline{j} - 5\underline{k}) = 5(0) + (-1)(1) + (1)(-5) = -1 - 5 = -6 \neq 0$$

Not perpendicular

$$\underline{v} \cdot \underline{w} = (\underline{j} - 5\underline{k})(-15\underline{i} + 3\underline{j} - 3\underline{k}) = (0)(-15) + (1)(3) + (-5)(-3) = 3 + 15 = 18 \neq 0$$

Not perpendicular

$$\underline{u} \cdot \underline{w} = (5\underline{i} - \underline{j} + \underline{k}) \cdot (-15\underline{i} + 3\underline{j} - 3\underline{k}) = (5)(-15) + (-1)(3) + (1)(-3) = -75 - 3 - 3 = -81 \neq 0$$

Not perpendicular

Now, $\underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k} = -3(5\underline{i} - \underline{j} + \underline{k}) = -3\underline{u}$

So, $\underline{u}$ & $\underline{w}$ are parallel.

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ii. $\underline{u}=\underline{i}+2\underline{j}-\underline{k};\underline{v}=-\underline{i}+\underline{j}+\underline{k};\underline{w}=\frac{-\pi}{2}\underline{i}-\pi\underline{j}+\frac{\pi}{2}\underline{k}$

Sol: $\underline{u}=\underline{i}+2\underline{j}-\underline{k};\underline{v}=-\underline{i}+\underline{j}+\underline{k};\underline{w}=\frac{-\pi}{2}\underline{i}-\pi\underline{j}+\frac{\pi}{2}\underline{k}$

For parallel,

$$\underline{w}=\frac{-\pi}{2}\underline{i}-\pi\underline{j}+\frac{\pi}{2}\underline{k}=\frac{-\pi\underline{i}-2\pi\underline{j}+\pi\underline{k}}{2}=\frac{-\pi}{2}\left(\underline{i}+2\underline{j}-\underline{k}\right)$$

$$\underline{w}=\frac{-\pi}{2}\underline{u}$$

So, $\underline{u}$ & $\underline{w}$ are parallel.

Now for perpendicular

$$\underline{u}\cdot\underline{v}=\left(\underline{i}+2\underline{j}-\underline{k}\right)\left(-\underline{i}+\underline{j}+\underline{k}\right)=1(-1)+2(1)+(-1)(1)=-1+2-1=0$$

For $\underline{u}$ & $\underline{w}$ no need to check for perpendicular because they are parallel.

Q7. **Use the definition of cross product, for any vectors $\underline{u},\underline{v},\underline{w}$ and scalar k, prove that**

i. $\underline{u}\times(-\underline{u})=0$ **ii.** $\underline{u}\times\underline{v}=-\underline{v}\times\underline{u}$

iii. $\underline{u}\times(k\underline{v})=(k\underline{u})\times\underline{v}=k(\underline{u}\times\underline{v})$ **iv.** $\underline{u}\times(\underline{v}+\underline{w})=(\underline{u}\times\underline{v})+(\underline{u}\times\underline{w})$

Sol: Let $\underline{u}=(u_1,u_2,u_3),\underline{v}=(v_1,v_2,v_3),\underline{w}=(w_1,w_2,w_3)$.

i. $\underline{u}\times(-\underline{u})=0$

Sol: $\underline{u}\times(-\underline{u})=0$

$$\underline{u}\times(-\underline{u})=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\u_1&u_2&u_3\\-u_1&-u_2&-u_3\end{vmatrix}$$

We can factor out -1 from the third row:

$$=(-1)\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\u_1&u_2&u_3\\u_1&u_2&u_3\end{vmatrix}\quad(R_2 \text{ and } R_3 \text{ are identical})$$

$$=(-1)\cdot 0=0$$

Thus, $\underline{u}\times(-\underline{u})=0$.

ii. $\underline{u}\times\underline{v}=\underline{v}\times\underline{u}$

Sol: $\underline{u}\times\underline{v}=\underline{v}\times\underline{u}$

Now, consider $\underline{v}\times\underline{u}$:

$$\underline{v}\times\underline{u}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\v_1&v_2&v_3\\u_1&u_2&u_3\end{vmatrix}$$

$$\underline{v}\times\underline{u}=-\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\u_1&u_2&u_3\\v_1&v_2&v_3\end{vmatrix}\left(interchanging\ R_2 \text{ and } R_3\right)$$

$$\underline{v}\times\underline{u}=-\underline{u}\times\underline{v}$$

Therefore, $\underline{u}\times\underline{v}=-\underline{v}\times\underline{u}$.

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iii. $\underline{u}\times(k\underline{v})=(k\underline{u})\times\underline{v}=k(\underline{u}\times\underline{v})$

Sol: $\underline{u}\times(k\underline{v})=(k\underline{u})\times\underline{v}=k(\underline{u}\times\underline{v})$

Let's start with $\underline{u}\times(k\underline{v})$:

$$\underline{u}\times(k\underline{v})=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ u_1 & u_2 & u_3\\ kv_1 & kv_2 & kv_3\end{vmatrix}$$

A property of determinants is that if a row (or column) is multiplied by a scalar k, the determinant is multiplied by k. We can factor k out of the third row:

$$=k\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{vmatrix}=k(\underline{u}\times\underline{v})$$

Now, let's consider $(k\underline{u})\times\underline{v}$:

$$(k\underline{u})\times\underline{v}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ ku_1 & ku_2 & ku_3\\ v_1 & v_2 & v_3\end{vmatrix}$$

Similarly, we can factor k out of the second row:

$$=k\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{vmatrix}=k(\underline{u}\times\underline{v})$$

Thus, $\underline{u}\times(k\underline{v})=(k\underline{u})\times\underline{v}=k(\underline{u}\times\underline{v})$

iv. $\underline{u}\times(\underline{v}+\underline{w})=(\underline{u}\times\underline{v})+(\underline{u}\times\underline{w})$

Sol: $\underline{u}\times(\underline{v}+\underline{w})=(\underline{u}\times\underline{v})+(\underline{u}\times\underline{w})$

$$\underline{u}\times(\underline{v}+\underline{w})=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ u_1 & u_2 & u_3\\ v_1+w_1 & v_2+w_2 & v_3+w_3\end{vmatrix}$$

$$=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{vmatrix}+\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ u_1 & u_2 & u_3\\ w_1 & w_2 & w_3\end{vmatrix}$$ (Separate R_3 elements)

$$=(\underline{u}\times\underline{v})+(\underline{u}\times\underline{w})$$

Therefore, $\underline{u}\times(\underline{v}+\underline{w})=(\underline{u}\times\underline{v})+(\underline{u}\times\underline{w})$

Q8. **Prove that:** $\underline{a}\times(\underline{b}+\underline{c})+\underline{b}\times(\underline{c}+\underline{a})+\underline{c}\times(\underline{a}+\underline{b})=0$

Sol: L.H.S $=\underline{a}\times(\underline{b}+\underline{c})+\underline{b}\times(\underline{c}+\underline{a})+\underline{c}\times(\underline{a}+\underline{b})$

$=\underline{a}\times\underline{b}+\underline{a}\times\underline{c}+\underline{b}\times\underline{c}+\underline{b}\times\underline{a}+\underline{c}\times\underline{a}+\underline{c}\times\underline{b}$

$=\cancel{\underline{a}\times\underline{b}}+\cancel{\underline{a}\times\underline{c}}+\cancel{\underline{b}\times\underline{c}}-\cancel{\underline{a}\times\underline{b}}-\cancel{\underline{a}\times\underline{c}}-\cancel{\underline{b}\times\underline{c}}=0$ $\quad\underline{a}\times\underline{a}=0$ *and* $\underline{b}\times\underline{a}=-\underline{a}\times\underline{b}$

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Q9. If $\underline{a}+\underline{b}+\underline{c}=0$, then prove that $\underline{a}\times\underline{b}=\underline{b}\times\underline{c}=\underline{c}\times\underline{a}$

Sol: Take cross product with $\underline{a}$

$\underline{a}\times(\underline{a}+\underline{b}+\underline{c})=\underline{a}\times 0$

$\underline{a}\times\underline{a}+\underline{a}\times\underline{b}+\underline{a}\times\underline{c}=0$

$0+\underline{a}\times\underline{b}+\underline{a}\times\underline{c}=0$

$\underline{a}\times\underline{b}=-\underline{a}\times\underline{c}\Rightarrow\underline{a}\times\underline{b}=\underline{c}\times\underline{a}$ (i)

Take cross product with $\underline{b}$

$\underline{b}\times(\underline{a}+\underline{b}+\underline{c})=\underline{b}\times 0$

$\underline{b}\times\underline{a}+\underline{b}\times\underline{b}+\underline{b}\times\underline{c}=0$

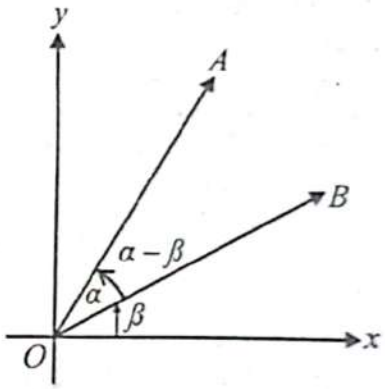
$\underline{b}\times\underline{a}+0+\underline{b}\times\underline{c}=0$

$-\underline{a}\times\underline{b}+\underline{b}\times\underline{c}=0\Rightarrow\underline{a}\times\underline{b}=\underline{b}\times\underline{c}$ (ii)

Combining (i) and (ii) then $\underline{a}\times\underline{b}=\underline{b}\times\underline{c}=\underline{c}\times\underline{a}$

Q10. Prove that: $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$ **Note:** Some Statement is incorrect in Book

Sol: Suppose two unit vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$



$\overrightarrow{OA}=\cos\alpha\underline{i}+\sin\alpha\underline{j}$

$\overrightarrow{OB}=\cos\beta\underline{i}+\sin\beta\underline{j}$

We know that $\overrightarrow{OB}\times\overrightarrow{OA}=\left|\overrightarrow{OB}\right|\left|\overrightarrow{OA}\right|\sin(\alpha-\beta)\underline{k}$

$$\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\ \cos\beta&\sin\beta&0\\ \cos\alpha&\sin\alpha&0\end{vmatrix}=1.1\sin(\alpha+\beta)\underline{k},\quad\left(\left|\overrightarrow{OA}\right|=\left|\overrightarrow{OB}\right|=1\right)$$

$\underline{i}(0-0)-\underline{j}(0-0)+\underline{k}(\sin\alpha\cos\beta-\cos\alpha\sin\beta)=\sin(\alpha-\beta)\underline{k}$

$(\sin\alpha\cos\beta-\cos\alpha\sin\beta)\underline{k}=\sin(\alpha-\beta)\underline{k}$

Hence $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$

Q11. Show that $|\underline{a}\times\underline{b}|^2=|\underline{a}|^2|\underline{b}|^2-(\underline{a}\cdot\underline{b})^2$

Sol: L.H.S $=|\underline{a}\times\underline{b}|^2=|\underline{a}|^2|\underline{b}|^2\sin^2\theta$ $\because\underline{a}\times\underline{b}=\underline{a}\underline{b}\sin\theta$

$=|\underline{a}|^2|\underline{b}|^2(1-\cos^2\theta)=|\underline{a}|^2|\underline{b}|^2-|\underline{a}|^2|\underline{b}|^2\cos^2\theta$

$=|\underline{a}|^2|\underline{b}|^2-(\underline{a}\cdot\underline{b})^2=R.H.S$ $\because\underline{a}\cdot\underline{b}=\underline{a}\underline{b}\cos\theta$

Q12. Use the definition of cross product, prove that for any vectors $\underline{u}$ and $\underline{v}$ $(\underline{u}+\underline{v})\times(\underline{u}-\underline{v})=-2(\underline{u}\times\underline{v})$

Sol: L.H.S $=(\underline{u}+\underline{v})\times(\underline{u}-\underline{v})=\underline{u}\times(\underline{u}-\underline{v})+\underline{v}\times(\underline{u}-\underline{v})$

$=(\underline{u}\times\underline{u})-(\underline{u}\times\underline{v})+(\underline{v}\times\underline{u})-(\underline{v}\times\underline{v})=0-(\underline{u}\times\underline{v})+(\underline{v}\times\underline{u})-0$

$=-(\underline{u}\times\underline{v})+(\underline{v}\times\underline{u})=-(\underline{u}\times\underline{v})+\left(-(\underline{u}\times\underline{v})\right)=-(\underline{u}\times\underline{v})-(\underline{u}\times\underline{v})=-2(\underline{u}\times\underline{v})$ = R.H.S

Q13. Find the moment about the point $M(1,-3,3)$ of the force represented by $\overrightarrow{AB}$, where the coordinates of points $A(4,3,-1)$ and $B(-1,3,7)$ are given.

Sol: $\underline{F}=(-1-4)\underline{i}+(3-3)\underline{j}+(7-(-1))\underline{k}=-5\underline{i}+0\underline{j}+8\underline{k}$

$\underline{r}=(4-1)\underline{i}+(3-(-3))\underline{j}+(-1-(3))\underline{k}=3\underline{i}+6\underline{j}-4\underline{k}$

$$\text{Moment}=\underline{r}\times\underline{F}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\ 3&6&-4\\ -5&0&8\end{vmatrix}=\underline{i}(48-0)-\underline{j}(24-20)+\underline{k}(0-(-30))=48\underline{i}-4\underline{j}+30\underline{k}$$

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Q14. A force $\underline{F}=6\underline{i}+4\underline{j}-4\underline{k}$ is applied at the point $A(1,-1,2)$. Find the moment of the force about the point $B(3,-2,3)$.

Sol: $\underline{F}=6\underline{i}+4\underline{j}-4\underline{k}$

$\underline{r}=(1-3)\underline{i}+(-1-(-2))\underline{j}+(2-3)\underline{k}=-2\underline{i}+1\underline{j}-1\underline{k}$

$$\text{Moment}=\underline{r}\times\underline{F}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\-2&1&-1\\6&4&-4\end{vmatrix}=\underline{i}(-4+4)-\underline{j}(8+6)+\underline{k}(-8-6)=0\underline{i}-14\underline{j}-14\underline{k}$$

Q15. Give a force $\underline{F}=2\underline{i}+\underline{j}-3\underline{k}$ acting at a point $A(1,-2,1)$. Find the moment of $\underline{F}$ about the point $B(2,0,-2)$.

Sol: $\underline{F}=2\underline{i}+1\underline{j}-3\underline{k}$

$\underline{r}=(1-2)\underline{i}+(-2-0)\underline{j}+(1-(-2))\underline{k}=-1\underline{i}-2\underline{j}+3\underline{k}$

$$\text{Moment}=\underline{r}\times\underline{F}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\-1&-2&3\\2&1&-3\end{vmatrix}=\underline{i}(6-3)-\underline{j}(3-6)+\underline{k}(-1-(-4))=3\underline{i}-(-3)\underline{j}+3\underline{k}$$

$M=3\underline{i}+3\underline{j}+3\underline{k}$

Q16. A force $\underline{F}=-2\underline{i}+\underline{j}-3\underline{k}$ is applied at $P(-1,-3,2)$. Find its moment about the point $Q(4,2,2)$.

Sol: $\underline{F}=-2\underline{i}+1\underline{j}-3\underline{k}$

$\underline{r}=(-1-4)\underline{i}+(-3-2)\underline{j}+(2-2)\underline{k}=-5\underline{i}-5\underline{j}+0\underline{k}$

$$\text{Moment}=\underline{r}\times\underline{F}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\-5&-5&0\\-2&1&-3\end{vmatrix}=\underline{i}(15-0)-\underline{j}(15-0)+\underline{k}(-5-10)=15\underline{i}-15\underline{j}-15\underline{k}$$

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